Practice problems #15

1. Let $y = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}$, $u_1 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$, $u_2 = \begin{bmatrix} -2/3 \\ 2/3 \\ 1/3 \end{bmatrix}$, and $W = \text{Span}\{u_1, u_2\}$.

   (a) Show that $\{u_1, u_2\}$ is an orthonormal basis of W.

   (b) Is $y \in W$? Why or why not?

   (c) Find the component of y orthogonal to both u_1 and u_2.

Answers or Hints are on page 2.

Answers or Hints:

#1(a): Easy.
#1(b): Is the equation $x_1u_1 + x_2u_2 = y$ consistent? Ans: No.
#1(c): $y = a + b$, where $a \in W$, and $b \in W^{\perp}$. Note that the component of y orthogonal to both u_1 and u_2 is b, and $b = y - a$ where $a = c_1u_1 + c_2u_2$. Show that $c_i = y \cdot u_i$ from the equation $y \cdot u_i = (a + b) \cdot u_i$ as we did in class.